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A tale of two mergers: Low back vowels in Baltimore

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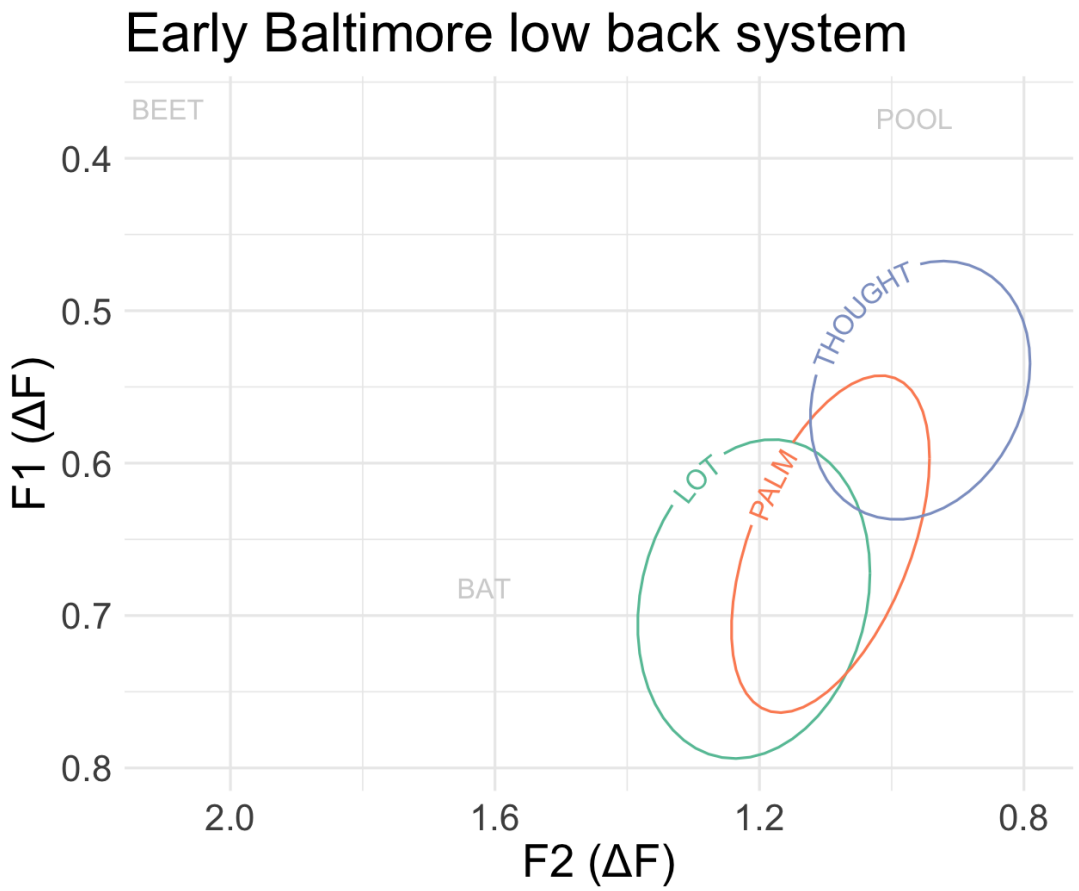
[3] Independent Scholar

Introduction

- In the early 20th century, Baltimore English had three low back vowels: short LOT and long PALM and THOUGHT
- Over the course of the 20th century, this system has been lost via two pathways:
  1. LOT-PALM merger (mainly for white speakers)
  2. PALM-THOUGHT merger (mainly for Black speakers)
- This talk addresses:
  - how these mergers took place
  - why white and Black Baltimoreans differ in patterns of merger

LOT and PALM

- Typically merged in rhotic dialects (Wells 1982)
- When distinct, PALM is longer than LOT (Kurath & McDavid 1961)
- Their relative positions vary (Kurath & McDavid 1961)



Types of mergers (Dinkin 2016)

- Merger by approximation: phonemes gradually approach each other phonetically
- Merger by (lexical) transfer: words are transferred from one phoneme to another one word at a time
- Merger by expansion: the contrast between two phonemes is abruptly lost
  - Attributed to contact phenomena (Herold 1997)
- Merger by phonological transfer: words are transferred from one phoneme to another one environment at a time

Data and methods

Baltimore Neighborhood Heritage Project (BNHP)

- Oral history interviews conducted 1978–1981 (Shopes 1981)
- Speakers born 1879–1951
- Current sample: 24 speakers born 1896–1940

Race	Gender	Count
Black	F	5
Black	M	3
White	F	8
White	M	8

Corpus of Linguistic Interviews in Charm City (CLICC)

- 48 sociolinguistic interviews conducted 2025–2026
- Elicitation tasks targeting low back vowels and pre-rhotic back vowels
- Speakers born 1940–2008
- Conversational data from 15 participants

Race	Gender	Count (all)	Count (convo.)
Asian	M	1	0
Black	F	8	3
Black	M	6	3
Hispanic	M	1	0
Biracial	F	1	0
Biracial	M	2	0
Biracial	NB	1	0
White	F	15	5
White	M	13	4

Data processing

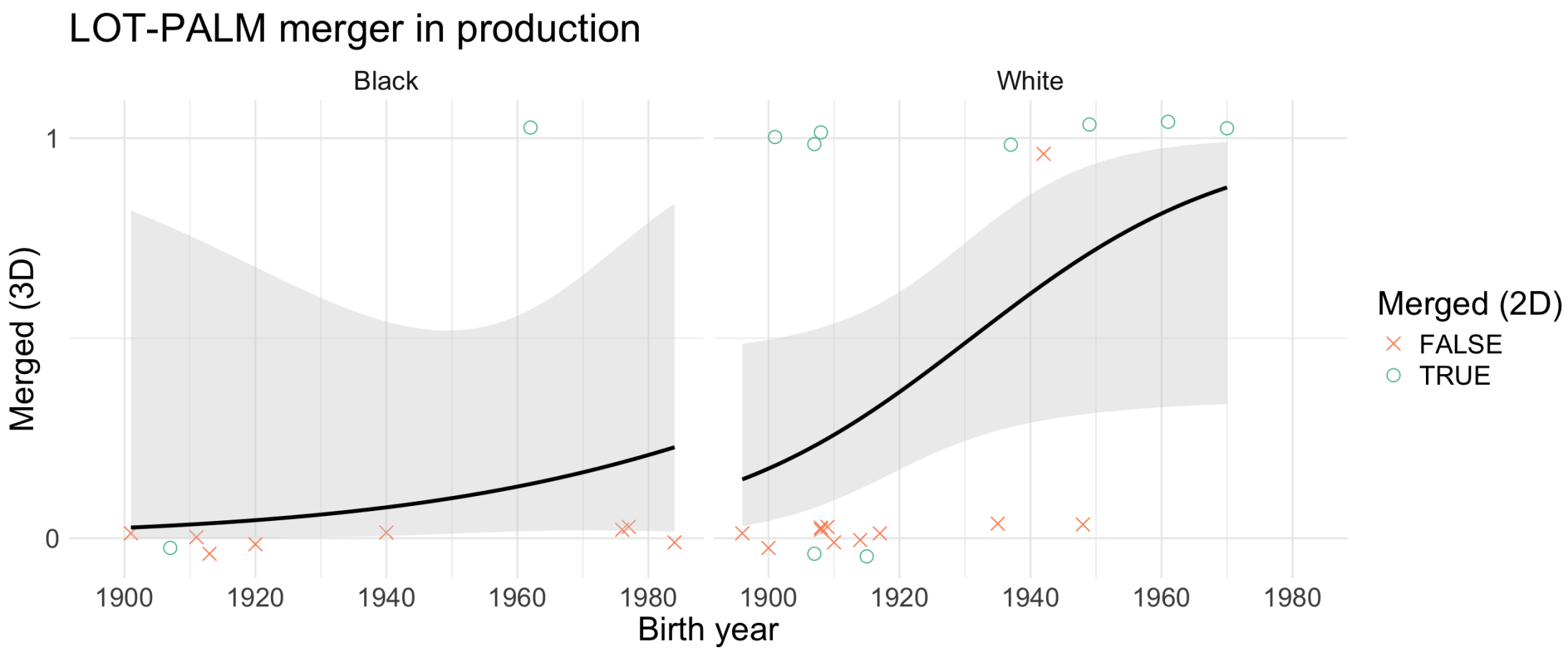
1. Transcription
2. Forced alignment with MFA (McAuliffe et al. 2017)
3. Formant extraction
    1. BNHP: with Darla (Reddy & Stanford 2015)
    2. CLICC: with New-Fave (Fruehwald & Maier 2026)
4. Divide by environment (see Stanley 2022)
5. Outlier removal (Leys et al. 2018)
6. ΔF normalization (Johnson 2020) in Tidynorm (Fruehwald 2025)

MANOVAs

- Multivariate analysis of variance used to assess merger in production (Nycz & Hall-Lew 2013)
  - Only use data CLICC data from the conversational portion
- Predictor variables:
  - Lexical set
  - Syllable type (open vs. closed)
  - Following voicing/place/manner
- Outcome variables: F1, F2, (log duration)
- Significant *p*-value (<0.05) for lexical set predictor taken as evidence for a distinction
  - Stanley & Sneller's (2023) not applicable with >2 outcome variables (Joseph A. Stanley, personal communication, 9 April 2026)

Documenting the loss of PALM

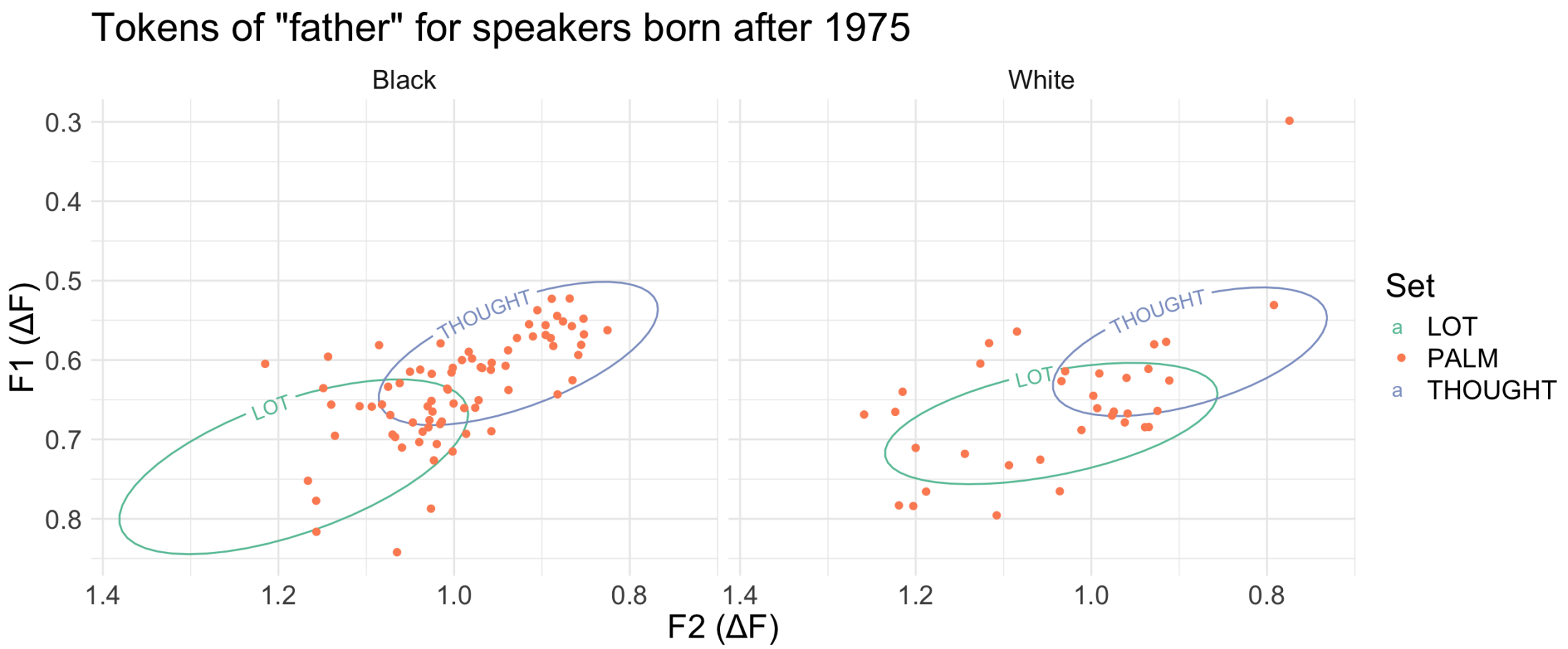
LOT-PALM merger in Baltimore



PALM-THOUGHT merger in Baltimore



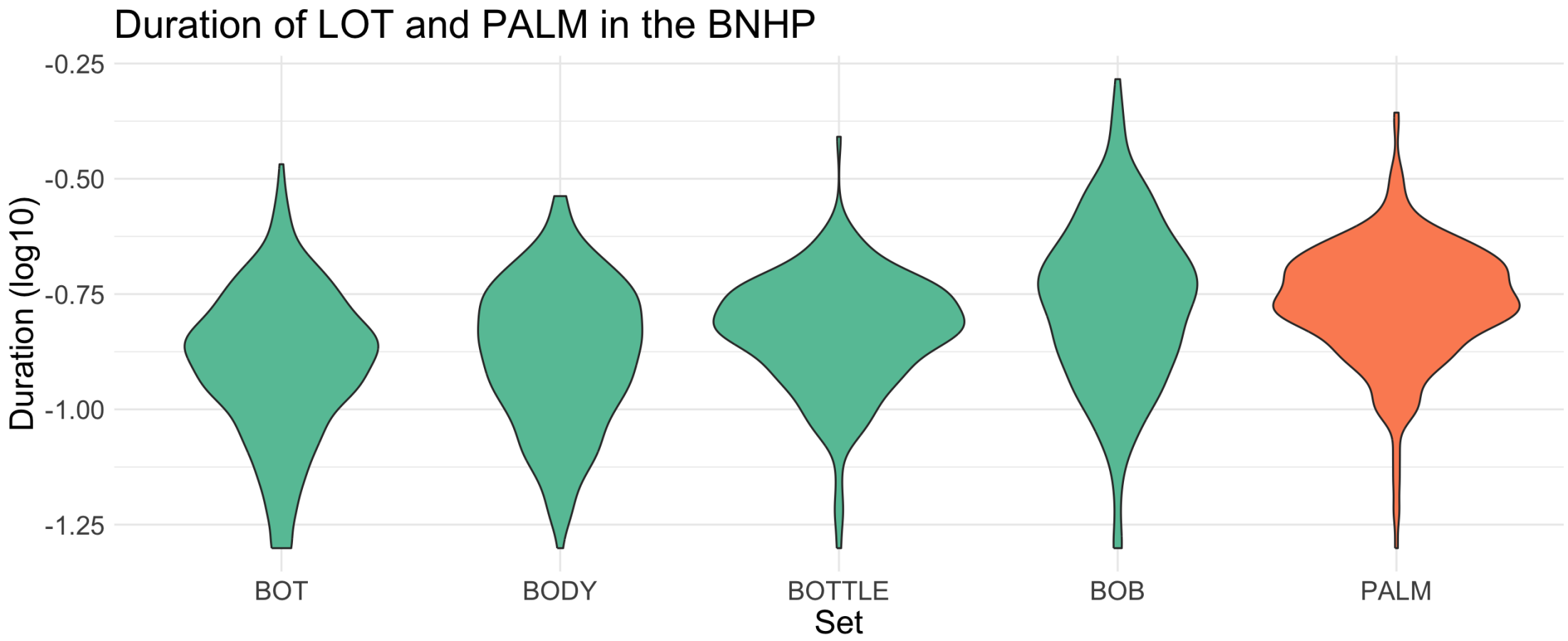
PALM for younger Baltimoreans



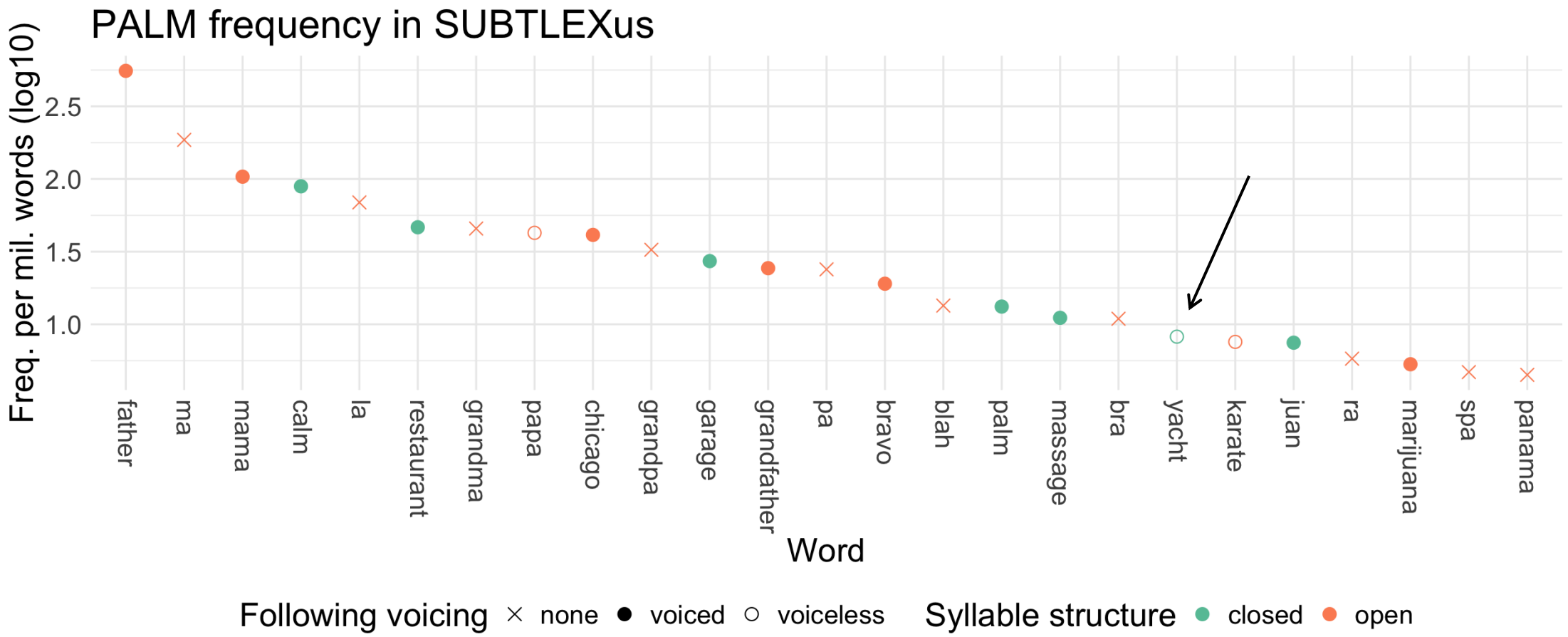
Tokens of LOT before voiced coda consonants excluded due to conditioned merger to PALM/THOUGHT

Explaining the loss of PALM

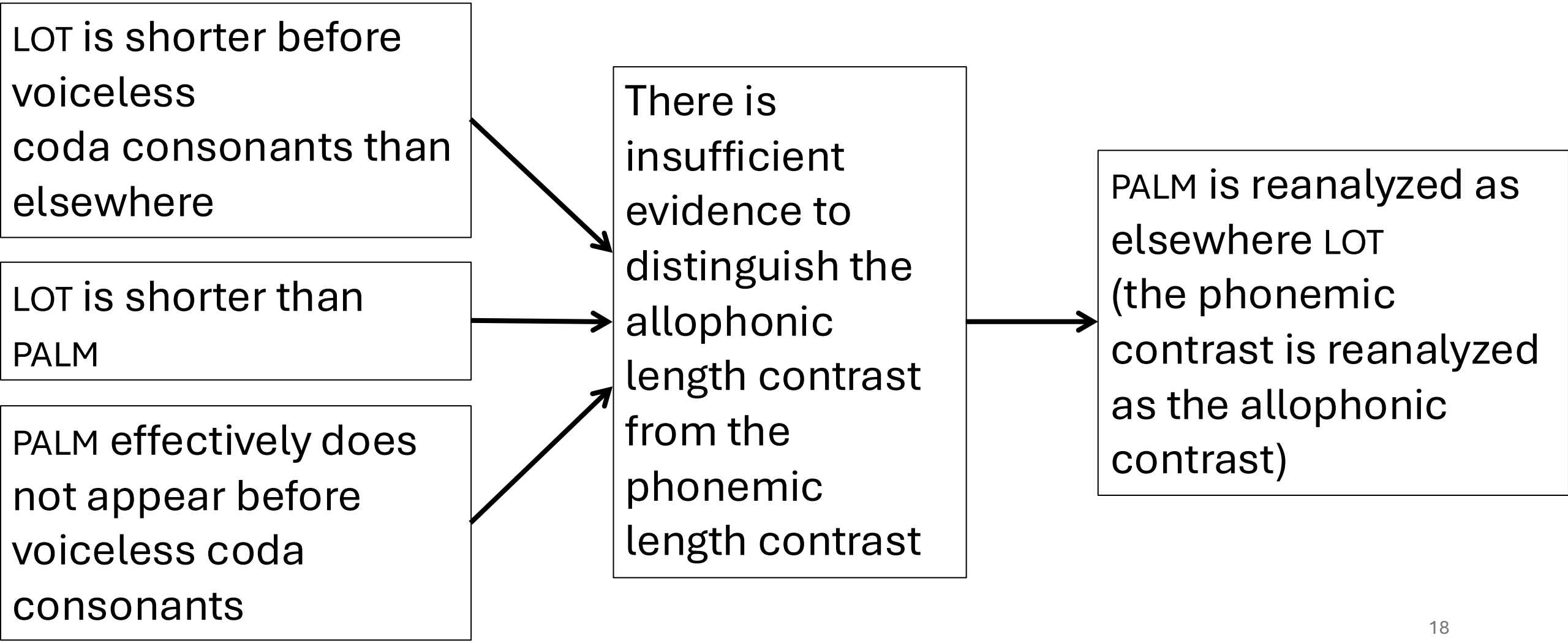
Allophonic and phonemic contrasts in duration



Marginal contrast



Merger via phonological reanalysis



Resisting LOT-PALM merger

- This explains how LOT-PALM merger arises
- But why do Black Baltimoreans resist LOT-PALM merger?
- This is due to the lower rate of rhoticity among Black Baltimoreans: non-rhoticity reinforces the LOT-PALM distinction (see also Labov 2006)

Non-rhoticity and PALM

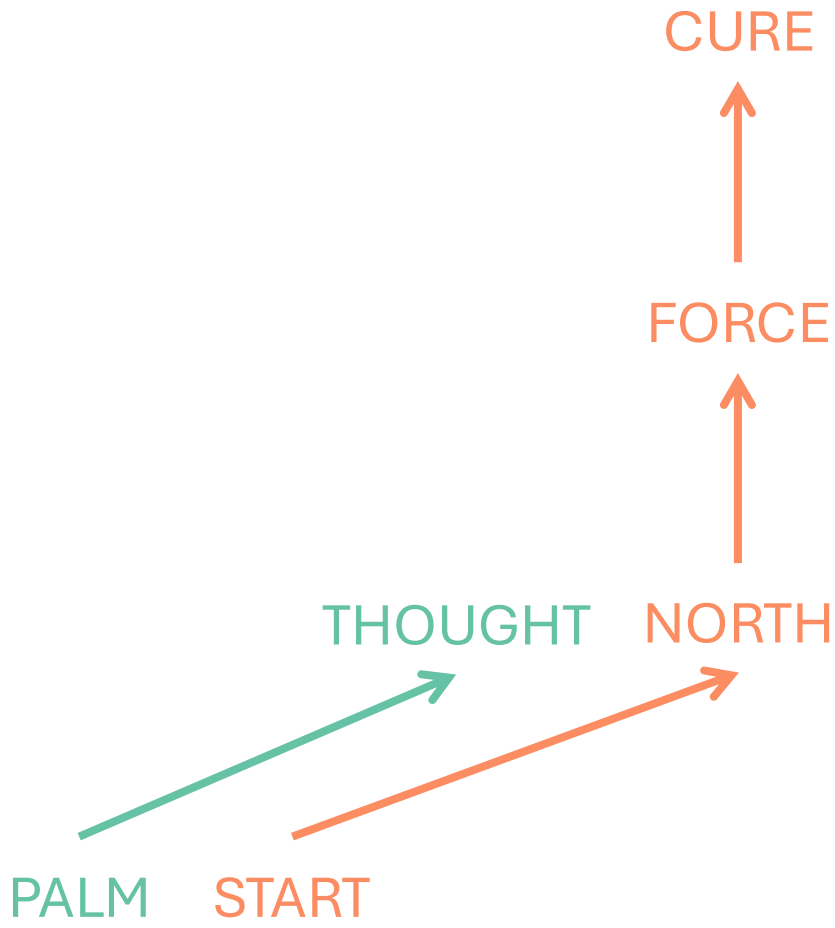
- START is normally pre-rhotic PALM > non-rhoticity in START enlarges the PALM set (Wells 1982)
- Increases functional load of LOT-PALM contrast
- Makes LOT-PALM contrast less marginal

	mark	*mock*
UF	/mɑːɹk/	/mɑk/
r-deletion	mɑːØk	---
SF	[mɑːk]	[mɑk]

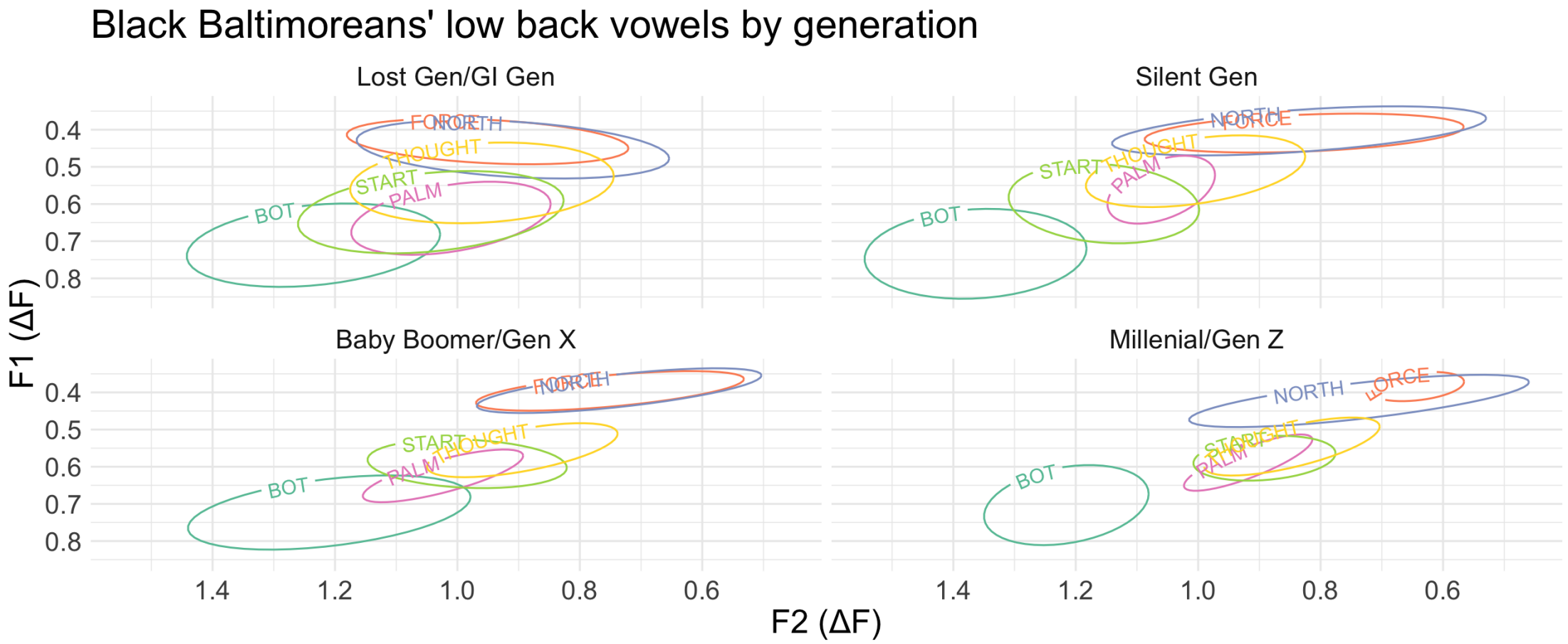
Generalizing the Back Vowel Chain Shift before /r/ (BVCS)

PALM instead raises as a generalization of the BVCS:

0. PALM remains distinct from LOT
1. START raises towards THOUGHT in the BVCS
2. START-raising generalizes to PALM
3. PALM merges with THOUGHT



START becomes pre-rhotic THOUGHT later for Black Baltimoreans



Summary

- For white Baltimoreans, the marginal LOT-PALM contrast is reanalyzed as allophonic (merger by expansion)
  - Can account for LOT-PALM merger in other rhotic varieties
- For Black Baltimoreans, non-rhoticity reinforces the LOT-PALM contrast, preventing merger
  - PALM eventually raises as a generalization of START-raising, leading to PALM-THOUGHT merger (by approximation)

Discussion

- Shows that merger by expansion can occur via phonological reanalysis rather than language contact (cf. Herold 1997)
- Shows that linguistic differences between racial groups can be internally motivated (by rhoticity)
  - Difference in rhoticity may be socially motivated (see Bonfiglio 2002), providing an **indirect social motivation**

Thank you!

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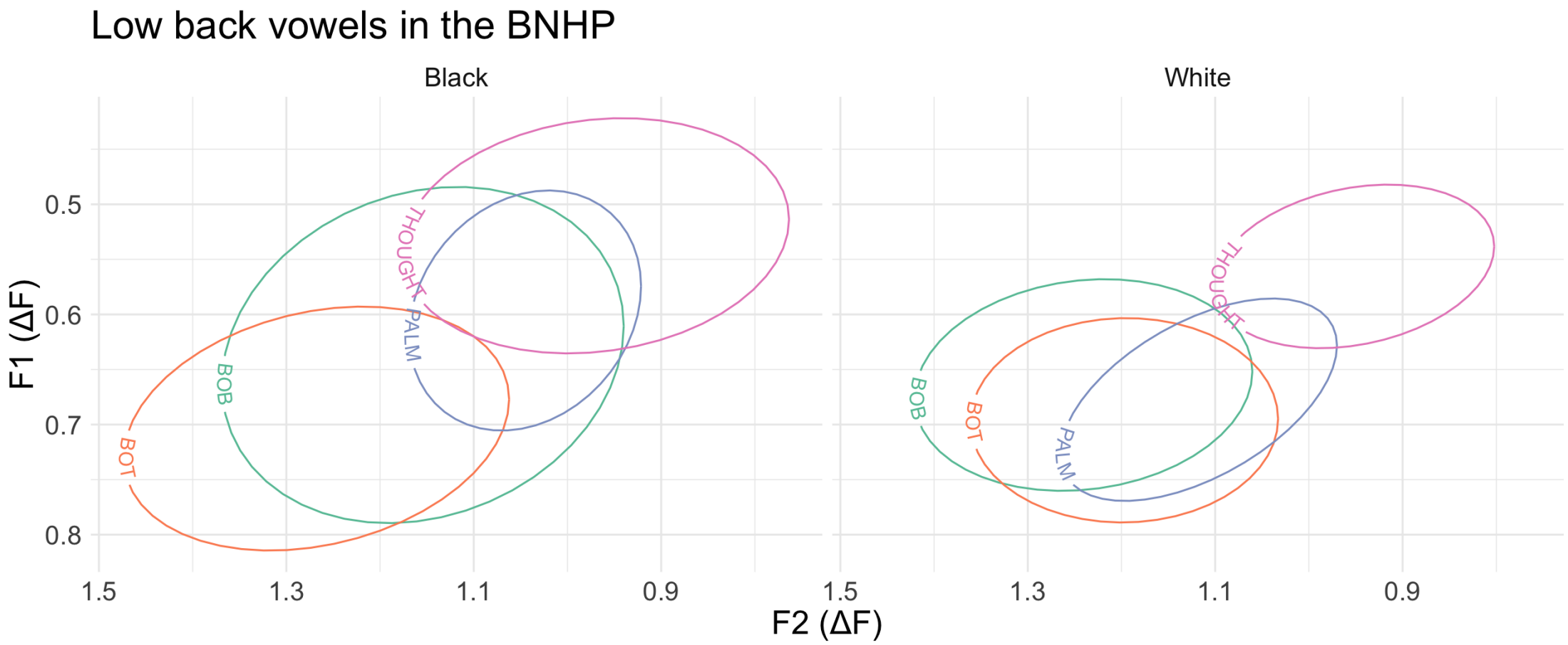
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Appendix A: Baltimore's low back vowels



BOT = LOT before voiceless coda consonants; BOB = LOT before voiced consonants

Appendix B: What about F2?

- PALM was backer than LOT in Baltimore
- Greater duration > more time to reach articulatory target
- F2 difference can be reanalyzed as a byproduct of the duration difference

